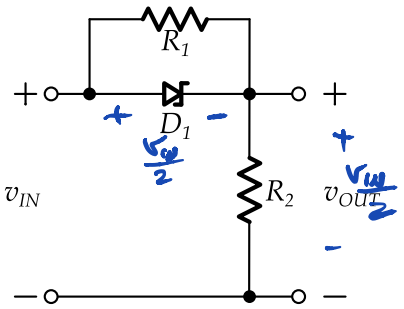
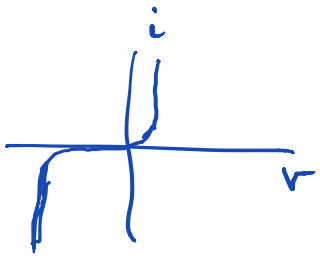
1. For the circuit below, assume $V_{D0} = 0.7$ volts, $V_{Z0} = 1.5$ volts, $R_1 = R_2$, and $r_D = r_Z = 0$.

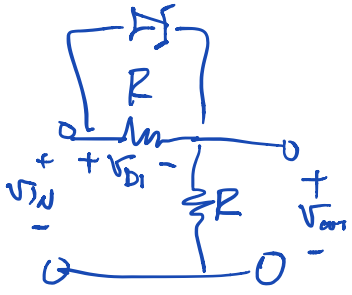




(a) (10 points) Write an expression for v_{OUT} in terms of v_{IN}.

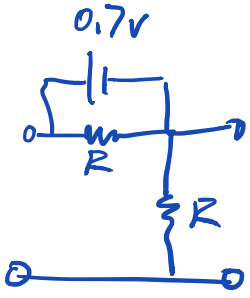
$$v_{OUT} = \begin{cases} v_{IN} - V_{D0} & v_{IN} \geq 2V_{D0} \\ \frac{1}{2} v_{IN} & -2V_{Z0} < v_{IN} < 2V_{D0} \\ v_{IN} + V_{Z0} & v_{IN} \leq -2V_{Z0} \end{cases}$$

$$= \begin{cases} v_{IN} - 0.7\text{V} & v_{IN} \geq 1.4\text{V} \\ \frac{1}{2} v_{IN} & -3\text{V} < v_{IN} < 1.4\text{V} \\ v_{IN} + 1.5\text{V} & v_{IN} \leq -3\text{V} \end{cases}$$

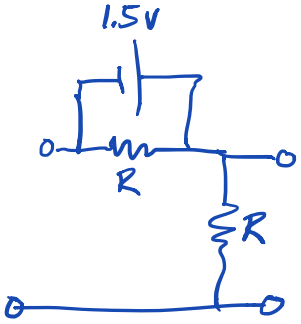


$v_{IN} = v_{D1} + v_{OUT}$

and $v_{D1} = v_{OUT} = \frac{v_{IN}}{2}$



$v_{OUT} = v_{IN} - 0.7\text{V}$

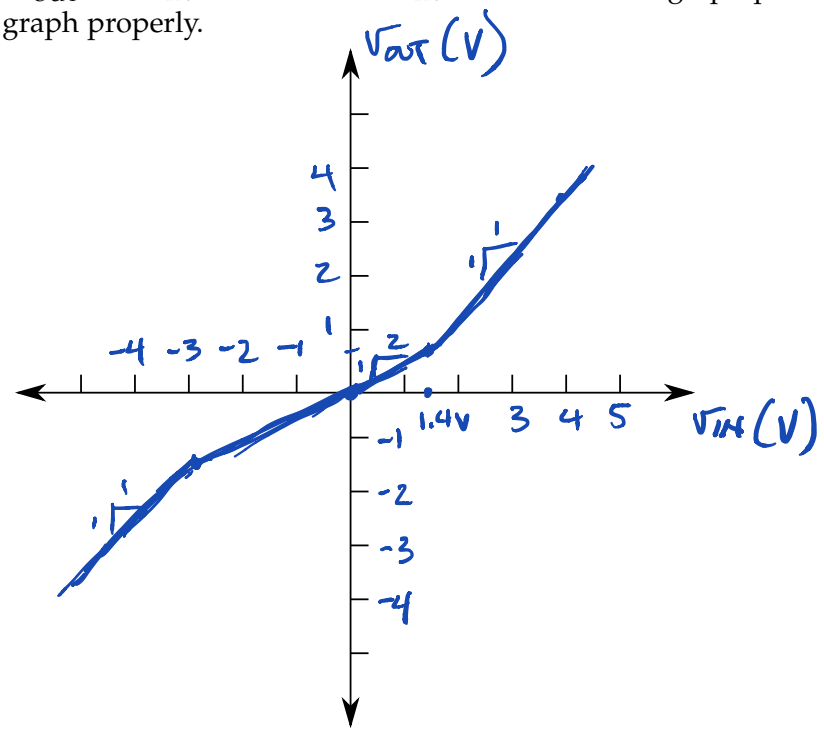


$v_{OUT} = v_{IN} + 1.5\text{V}$

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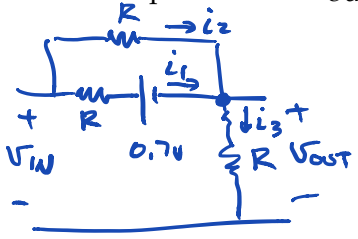
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(b) (5 points) Sketch v_{OUT} and v_{IN} for -4 volts $< v_{IN} < 4$ volts on the graph provided. Be sure to label the graph properly.



(c) (10 points) Now, replace the Zener diode (D_1) with a regular diode, and use $V_{D0} = 0.7$ volts and $r_D = R_1 = R_2$. Write an expression for v_{OUT} in terms of v_{IN}.

$v_{IN} > 2V_{D0}$:



$i_1 = (v_{IN} - V_{D0} - v_{OUT})/R$

$i_2 = (v_{IN} - v_{OUT})/R$

$i_3 = v_{OUT}/R$

since $i_3 = i_1 + i_2$

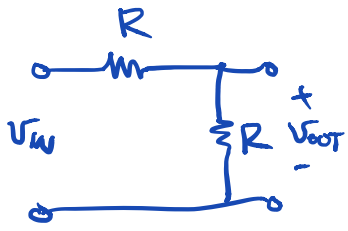
$\frac{v_{OUT}}{R} = \frac{v_{IN} - V_{D0} - v_{OUT}}{R} + \frac{v_{IN} - v_{OUT}}{R}$

$v_{OUT} = 2v_{IN} - 2v_{OUT} - V_{D0}$

$3v_{OUT} = 2v_{IN} - V_{D0}$

$v_{OUT} = \frac{2v_{IN} - V_{D0}}{3}$

$v_{IN} \leq 2V_{D0}$

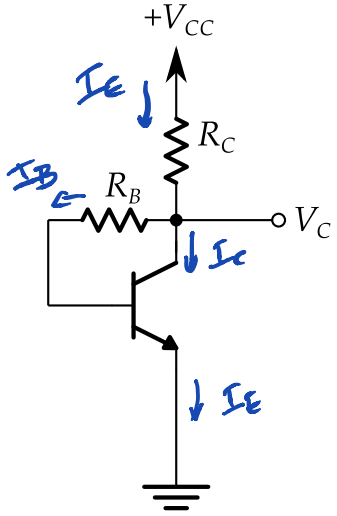


$$v_{OUT} = \begin{cases} \frac{1}{2} v_{IN} & v_{IN} \leq 2V_{D0} \\ \frac{2v_{IN} - V_{D0}}{3} & v_{IN} > 2V_{D0} \end{cases}$$

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2. Use the following circuit to complete this problem.



(a) (10 points) Write expressions for V_C and the currents in the transistor in terms of R_B, R_C, V_{CC}, and β.

$V_{cc} = I_E R_c + I_B R_B + V_{BE}$

$\Rightarrow I_B = \frac{V_{cc} - V_{BE}}{(\beta+1)R_c + R_B}$ $\quad I_E = (\beta+1) I_B$ $\quad I_c = \beta I_B$

$V_c = V_{cc} - I_E R_c$ or $V_c = V_{BE} + I_B R_B$

(b) (10 points) Find values for R_B and R_C so that $V_C = V_{CC}/2$ and $r_\pi = 1\,\text{k}\Omega$ when $V_{CC} = 25\,\text{V}$ and $\beta = 100$.

since $r_\pi = \frac{V_T}{I_B}$, $I_B = \frac{V_T}{r_\pi} = \frac{0.026V}{1K\Omega} = 0.026\,mA$

since $V_c = V_{BE} + I_B R_B \Rightarrow R_B = \frac{V_c - V_{BE}}{I_B} = \frac{12.5 - 0.7}{0.026} = 454\,K\Omega$

since $V_c = V_{cc} - I_E R_c \Rightarrow R_c = \frac{V_{cc} - V_c}{I_E} = \frac{25 - 12.5}{101 \cdot 0.026} = 4.76\,K\Omega$

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(c) (10 points) What is the range of V_C when $50 \leq \beta \leq 150$?

Now that we have values for R_B + R_C, we can use the equations from (a) to find V_C with the new β values

for $\beta = 50$ $\quad I_B = \dfrac{25 - 0.7}{51 \cdot 4.76 + 454} = 0.0349\,mA$

$V_c = 0.7 + (0.035)(454) = 16.5\ V$

for $\beta = 150$ $\quad I_B = \dfrac{25 - 0.7}{151 \cdot 4.76 + 454} = 0.0207\,mA$

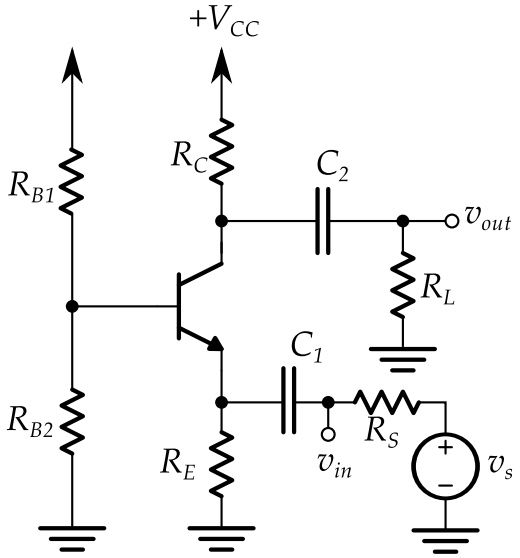
$V_c = 0.7 + (0.0207)(454) = 10.1\ V$

$10.1\ V < V_c < 16.5\ V$

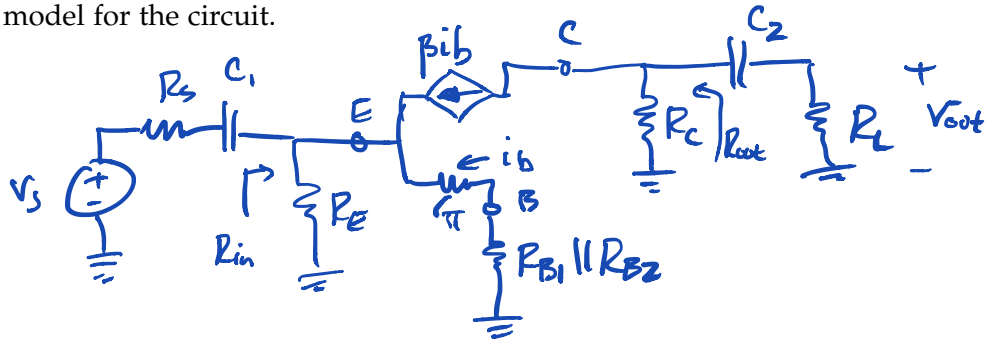
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3. Use the following circuit to complete this problem.



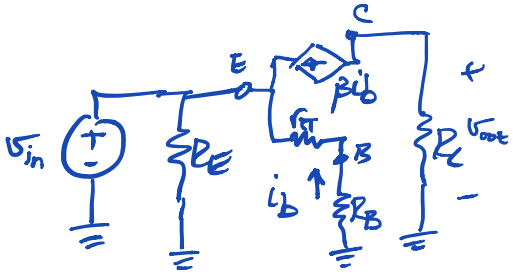
(a) (10 points) Assuming that the transistor is in an active mode, sketch the small signal model for the circuit.



(b) (10 points) Write an expression for the *midband gain* of the circuit, $\frac{v_{out}}{v_{in}}$, with the load and source removed. The solution should be expressed in terms of resistors, β, r_π.

$V_{out} = -\beta i_b \cdot R_C \qquad V_{in} = -i_b (r_\pi + R_{B1} \| R_{B2})$

$$\frac{V_{out}}{V_{in}} = \frac{\beta R_C}{r_\pi + R_{B1} \| R_{B2}}$$



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$$i_{in} = \frac{v_{in}}{R_E} - (\beta+1) i_b$$

$$v_{in} = -i_b (r_\pi + R_B)$$

$$R_{in} = \frac{v_{in}}{i_{in}} = \frac{-i_b (r_\pi + R_B)}{\frac{-i_b (r_\pi + R_B)}{R_E} - (\beta+1) i_b} = \frac{(r_\pi + R_B)}{\frac{r_\pi + R_B}{R_E} + (\beta+1)} = \frac{R_E \cdot \frac{r_\pi + R_B}{\beta+1}}{\frac{r_\pi + R_B}{\beta+1} + R_E} = R_E \| \left[\frac{r_\pi + R_B}{\beta+1} \right]$$

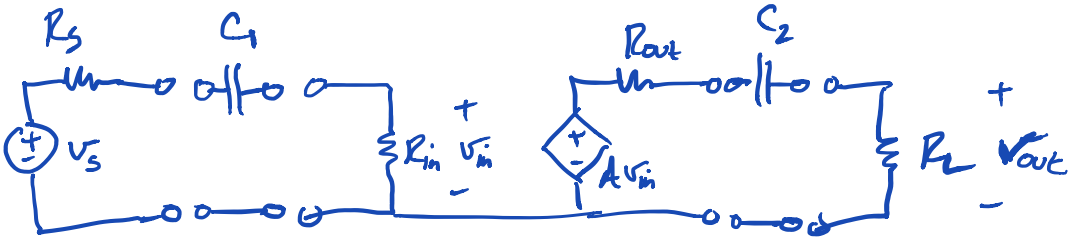
(c) (10 points) Find expression for the input and output resistance for the *midband* with the source and load removed.

$$R_{in} = R_E \| \frac{v_{in}}{-i_e} = R_E \| \left(\frac{v_{in}}{-(\beta+1) i_b} \right) = R_E \| \left[\frac{-i_b (r_\pi + R_{B1} \| R_{B2})}{-(\beta+1) i_b} \right]$$

$$= R_E \| \left[\frac{r_\pi + R_{B1} \| R_{B2}}{(\beta+1)} \right]$$

$$\boxed{R_{in} = R_E \| \left[\frac{r_\pi + R_{B1} \| R_{B2}}{(\beta+1)} \right]}$$

$$\boxed{R_{out} = R_L}$$



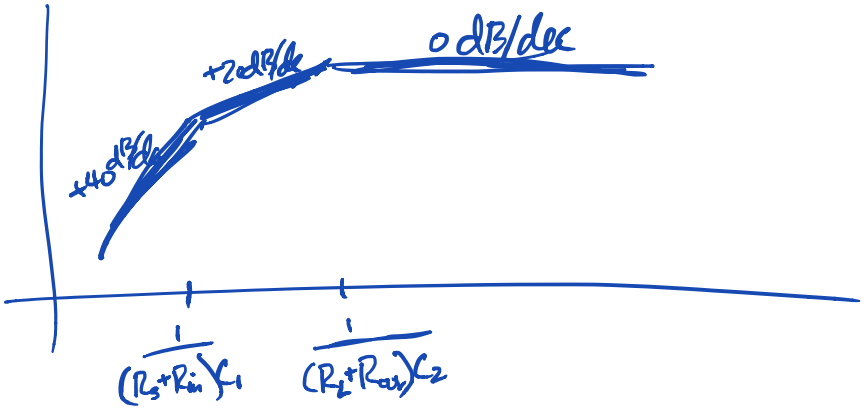
$$\bar{V}_{out} = A \bar{V}_{in} \cdot \frac{R_L}{R_{out} + R_L + \frac{1}{j\omega C_2}} = A \bar{V}_{in} \cdot \frac{j\omega R_L C_2}{1 + j\omega (R_{out} + R_L) C_2}$$

$$\bar{V}_{in} = \bar{V}_s \cdot \frac{R_{in}}{R_S + R_{in} + \frac{1}{j\omega C_1}} = \bar{V}_s \cdot \frac{j\omega R_{in} C_1}{1 + j\omega (R_S + R_{in}) C_1}$$

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(b) (5 points) Make a rough sketch of a possible Bode magnitude plot for this circuit.



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